

URLLC-Based Multi-Hop Communication Using DECT-2020 NR

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Abstract—DECT-2020 New Radio (NR) is a Radio Interface Technology (RIT) introduced by ETSI for massive Machine Type Communications (mMTC) and Ultra-Reliable Low Latency Communications (URLLC) use cases. A single DECT-2020 NR Radio Device (RD) in Fixed Termination (FT) mode announces radio resources for the connected RDs in Portable Termination (PT) mode. An RD can also operate in both modes simultaneously as an FT/PT. An RD in FT mode is responsible for packet routing for the connected radio devices. However, for multi-hop communication, RDs in FT mode flood the packet, which utilizes more network resources and introduces additional latency. In this paper, we propose a routing procedure for URLLC-based multi-hop communication using the MAC layer and convergence layer functionalities of DECT-2020 NR. Moreover, we formulate the optimization problem for optimal path selection based on user-plane latency, which is our previous contribution to the ETSI self-evaluation technical report of DECT-2020 NR and propose a graph theory-based solution. The investigations demonstrate that an improvement of up to 15% in throughput and a reduction of up to 15% in end-to-end latency is observed in comparison with the standard flooding procedure.

Index Terms—DECT-2020 NR, URLLC, Multi-Hop Communication, End-to-End Latency, Routing

I. INTRODUCTION

DECT-2020 NR standard is one of the RITs released by the European Telecommunications Standards Institute (ETSI) in 2020, which supports use cases of mMTC and URLLC following the IMT-2020 guidelines [1]. In DECT-2020 NR, the Radio Device (RD) is capable of operating in three distinct modes: Fixed Termination (FT) mode, Portable Termination (PT) mode, and a combined FT/PT mode. The RDs operating in these modes are denoted as RD_{FT} , RD_{PT} and $RD_{FT/PT}$ respectively. DECT-2020 NR introduces a downlink routing protocol for unicast, multicast, and broadcast communications, particularly in single-hop packet transmission scenarios. The gateway (i.e., connected with the external network) operating in FT mode manages the list of connected RDs and is responsible for packet routing. In single-hop transmissions, the destination RDs operating in PT mode are directly connected to the gateway, enabling direct packet forwarding. However, DECT-2020 NR lacks a routing mechanism for multi-hop communication. In such cases, the gateway attempts to locate the destination node; if unsuccessful, the gateway disseminates the packet to all connected RDs operating in $RD_{FT/PT}$ modes. Each receiving node reiterates the search for the destination node, and if the node remains undiscovered, the flooding process [2] is repeated until the packet reaches

TABLE I: Transmission configuration

Properties	Config1	Config2	Config3	Config4
(μ, β)	(1,1)	(2,1)	(4,1)	(8,1)
Bandwidth (MHz)	1.728	3.456	6.912	13.824
Symbol Length (μs)	41.667	20.833	10.417	5.208
Subcarrier Spacing (kHz)	27	54	108	216
Subslot Length (ms)	0.208	0.104	0.052	0.0260
Transmit Power (dBm)	23	23	23	23

its intended destination. In this paper, we propose a multi-hop routing procedure for DECT-2020 NR that ensures compliance with URLLC constraints, thereby mitigating the need for the flooding approach.

The existing literature review on DECT-2020 NR is mainly related to technology evaluation. For example, [3] conducted a link-level performance evaluation of a multi-hop assisted mMTC scenario for DECT-2020 NR. The performance was evaluated against the transmission configurations of DECT-2020 NR in terms of outage probability and bit error rate. [4] is another link-level performance evaluation of NOMA-aided multi-hop DECT-2020 NR for mMTC applications. The simulation results suggested that the NOMA-aided DECT-2020 NR multi-hop communication improves the outage probability and BER. [5] presented a link-level and system-level evaluation of DECT-2020 NR for the use case of URLLC and mMTC using the guidelines of IMT-2020. Authors in [6] evaluated the energy consumption of DECT-2020 NR network for battery-operated IoT devices and investigated the latencies with energy constraints. [7] presented an introduction to DECT-2020 NR and its applications in a 5G mMTC use case. The author presented a system-level evaluation and compared the investigations with existing LPWAN solutions in terms of the supported density of nodes. Author in [8] performed a performance evaluation of DECT-2020 NR in a real-life industrial environment for various RF bands. The investigation includes the evaluation of PER, BER, and RMS delay spread for a bandwidth of 13.824 MHz. [9] also presented a link-level simulation for the use case of URLLC and evaluated PER for various modulation schemes with HARQ implementation. The simulation results verified the requirements of URLLC, achieving link latency below 1ms and reliability of 99.99%. In this paper, we propose a multi-hop routing procedure leveraging the MAC and convergence layers of the DECT-2020 NR network to meet URLLC requirements. The gateway

maintains the network topology, with link costs defined by user plane latency to fulfill the DECT-2020 NR URLLC specifications as outlined in the standard [10] and ETSI evaluation report [11]. We formulate an optimization problem aimed at determining the optimal path selection to minimize end-to-end latency in multi-hop communication. Independently, the MAC and Convergence layers are employed to disseminate link costs to the gateway, which utilizes this information to construct a network graph. Leveraging the graph and the formulated optimization model, the gateway computes the optimal communication path using Dijkstra's algorithm. For downlink transmission, routing information is multiplexed in the MAC Service Data Unit (SDU) to guide packet forwarding, avoiding standard flooding. Simulation results show that our approach reduces end-to-end latency and improves throughput compared to the standard DECT-2020 NR flooding method. The rest of the paper is organized as follows: in section II the problem is formulated, in section III the proposed solution is presented, in section IV the performance is evaluated using simulation, and in section V concluding remarks are offered.

II. PROBLEM FORMULATION

We consider an indoor scenario of a DECT-2020 NR wireless network for the use case of URLLC communication. The network topology contains a gateway and radio devices operating in FT, FT/PT, and PT modes. Each radio device in FT or FT/PT mode shares the link cost (i.e., latency) with the gateway through uplink whenever there is a new association or re-association with other RDs as explained in section III. The gateway utilizes the shared information from the radio devices and creates a connectivity graph denoted as $G = (V, E)$. Where $V = \{v_1, v_2, v_3, \dots, v_n\}$ is the set of vertices (i.e., DECT-2020 NR nodes) and $E = \{e_1, e_2, e_3, \dots, e_n\}$ is the set of edges (links between the nodes).

The transmission model considers both the path loss model and the fading model to define received power at the nodes. The path loss model is chosen from the IMT-2020 guidelines [1] which is expressed as:

$$PL_{ij} = 16.9 \log_{10}(d_{ij}) + 32.8 + 20 \log_{10}(f_c) \quad (1)$$

where d_{ij} is the distance between the node i and node j , and f_c is the operating frequency. To consider a realistic channel feature, we used the Rayleigh fading channel model [12]. The probability density function of the Rayleigh fading channel model denoted as $f_X(x_{ij})$ is given by:

$$f_X(x_{ij}) = \frac{x_{ij}}{\sigma_{ij}^2} e^{-\frac{x_{ij}^2}{2\sigma_{ij}^2}}, \quad x_{ij} > 0 \quad (2)$$

where x_{ij} is the envelope of the received signal from node i to node j , and σ_{ij}^2 is the power of multi-path component. The expected value of the received power influenced by the fading channel is denoted as $E(x_{ij}^2)$. Thus, the total received

power from node i to node j including the effect of path loss and Rayleigh fading is expressed as:

$$P_r^{ij} = P_t^i + E(x_{ij}^2) - PL_{ij} \quad (3)$$

where P_t^i is the transmit power of a node i . To calculate the SNR from a node i to a node j , the following expression is used:

$$\gamma_{ij} = \frac{P_r^{ij}}{N} \geq \gamma_{th} \quad (4)$$

where N is the noise power and γ_{th} is the SNR threshold value which is calculated using our previous work [9]. The SNR threshold value corresponds to the packet error rate to achieve the reliability requirement of URLLC. The maximum achievable data rate for the link (i, j) is calculated using the Shannon formula given in equation (5).

$$R_{ij} = B_{ij} \log_2(1 + \gamma_{ij}) \quad (5)$$

where B_{ij} is the bandwidth for the link (i, j) . Based on the maximum achievable data rate and the data rate requirement of the application, a suitable Modulation and Coding Scheme (MCS) [10] is selected for the link (i, j) . DECT-2020 NR [10] supports various transmission configurations (μ, β) as shown in Table I.

The number of sub-slots required to transmit a packet is calculated by the following expression:

$$N_{\text{sub}} = \left\lceil \frac{\frac{S \cdot 8}{R_{ij}}}{T_{\text{sub}}^{ij}} \right\rceil \quad (6)$$

where S is the size of the packet in bytes, R_{ij} is the throughput of the link (i, j) with unit bits per second (bps) calculated by the selected MCS, and T_{sub}^{ij} is the subslot length in seconds given by the supported transmission configuration (μ, β) of the DECT-2020 NR standard [10]. Thus, the data packet transmission duration is given by:

$$t_{\text{dur}} = N_{\text{sub}} \cdot T_{\text{sub}}^{ij} \quad (7)$$

The total user-plane latency [11] of a link (i, j) for the frame-based system operation without Hybrid Automatic Repeat request (HARQ) is expressed in equation (8).

$$T_{ij}^N = t_{\text{sym}} + t_{\text{rd_tx}} + t_{\text{FA_switch}} + t_{\text{dur}} + t_{\text{rd_rx}} \quad (8)$$

where t_{sym} is the symbol alignment time, $t_{\text{rd_tx}}$ is the transmitter (TX) side processing delay, $t_{\text{FA_switch}}$ is the TX alignment and TDD switching time, and $t_{\text{rd_rx}}$ is the receiver (RX) side processing delay. The total user-plane latency with HARQ is calculated using the equation (9):

$$T_{ij}^H = T_{ij}^N + n \cdot [T_1 + T_2] \quad (9)$$

where n is the total number of re-transmissions. T_1 and T_2 are calculated by the following expressions.

$$T_1 = t_{\text{tx}} + t_{\text{FA_switch}} + t_{\text{HARQ}} + t_{\text{rd_rx}} \quad (10)$$

$$T_2 = t_{\text{rd_tx}} + t_{\text{FA_switch}} + t_{\text{dur}} + t_{\text{rd_rx}} \quad (11)$$

where T_1 represents HARQ feedback transmission delay and T_2 represents data re-transmission delay. In equation (10), t_{tx} is the TX side processing delay for HARQ feedback transmission, t_{HARQ} is the HARQ ACK/NACK transmission delay, t_{FA_switch} is the TX alignment and TDD switching time for HARQ ACK/NACK, and t_{rd-tx} is the receiver side processing delay for ACK/NACK. The symbols in equation (11) are explained in equation (8). The user plane latency expressed in equation (8) and (9) ensures the link-level latency requirement of URLLC communication, i.e., less than 1ms. The objective is to find a path from the gateway to a destination node such that the end-to-end latency for multi-hop communication is minimized. The optimization problem is thus formulated as:

$$\begin{aligned}
 & \text{minimize} \sum_{(i,j) \in E} \sum_{f \in F} X_{ij}^f T_{ij}^f (N_i^f + 1), \quad \forall f \in F \\
 & s.t. \\
 & \sum_{(i,j) \in E} X_{ij}^f = 1, \quad i \in S, \forall f \in F \\
 & \sum_{(i,j) \in E} X_{ij}^f = 1, \quad j \in D, \forall f \in F \\
 & \sum_{(i,j) \in E} X_{ij}^f = \sum_{(j,u) \in E} X_{ju}^f \quad \forall j \in V - \{S, D\}, \forall f \in F \\
 & c_{ij}^f \geq X_{ij}^f \quad \forall (i,j) \in E, \quad \forall f \in F \\
 & X_{ij}^f \in \{0, 1\} \quad \forall (i,j) \in E, \quad \forall f \in F
 \end{aligned} \tag{12}$$

where E is the set of edges in the network, f is a data flow which is defined as a set of packets required to transmit from the gateway/source node to a destination node, and F is the set of all data flows. X_{ij}^f is the binary variable of the optimization problem which represents whether a link (i, j) is selected for a flow f or not. If the link (i, j) is selected for transmission of flow f , the value of X_{ij}^f is set to 1; otherwise, the value of X_{ij}^f is set to 0. T_{ij}^f is the total latency of packet transmission on a link (i, j) for a flow f which is calculated using the equation (9). If the network does not require HARQ, the value of n in equation (9) is set to 0. N_i^f is the number of queueing data packets in a data flow f waiting to be transmitted at node i for the destination node. The constraints in equation (12) ensure that there should be one neighbour node connected to the source node S (i.e., gateway) to transmit data packets for a flow f . Similarly, for the destination node D , there should be one node connected to receive the data packets for the flow f . The other constraints ensure the loop-free intermediate links to deliver the packets successfully to the destination. The existence of links in the network is ensured by the binary constant $c_{ij}^f \in \{0, 1\}$. The proposed solution is presented in the section III in detail.

III. PROPOSED SOLUTION

In this research work, we propose a routing procedure for multi-hop communication using DECT-2020 NR while maintaining the URLLC constraints. In our considered scenario,

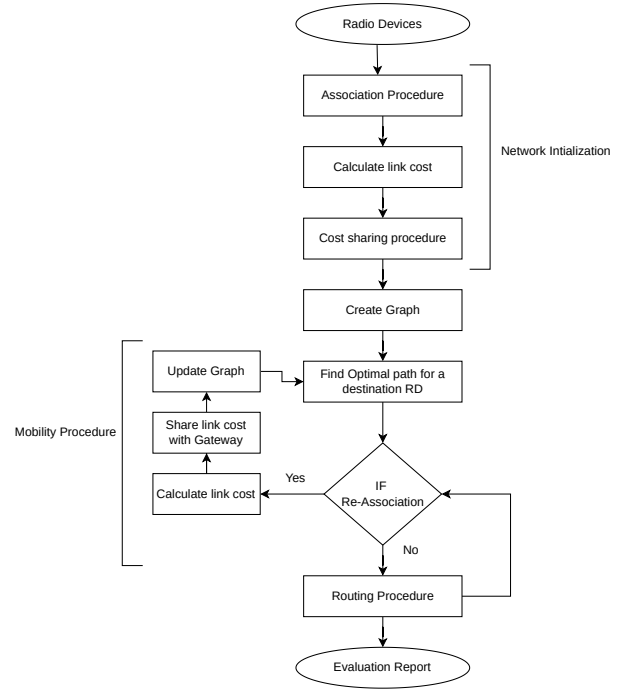


Fig. 1: Flow chart of proposed solution

we deployed one RD in FT mode on a static position as a gateway, some RDs in FT/PT mode are deployed on a fixed position to provide coverage for the region of interest, and the rest of the RDs in PT mode are mobile following a pre-defined path, e.g., AGVs in factory hall. The RD_{PT} can associate with more than one RD_{FT}/RD_{FT/PT} based on the minimum SNR requirement of URLLC communication. Moreover, if RD_{PT} changes its association, the respective RD_{FT/PT} sends the updated information to the gateway. For downlink communication, i.e., gateway to RD_{PT}, the gateway adds the routing information inside the packet and the gateway forwards the packet to the next hop. Using the embedded routing information, each intermediate hop forwards the packet to the next hop only, and eventually, the packet reaches the destination RD_{PT} following the optimal path. The flow chart of the proposed solution is shown in Fig. 1. Moreover, detailed descriptions of the

TABLE II: DECT-2020 NR Edges

Sr. No	Link	Cost
1	(RD _G , RD _{ID₁})	cost ₁
2	(RD _G , RD _{ID₂})	cost ₂
3	(RD _{ID₂} , RD _{ID₃})	cost ₃
4	(RD _{ID₂} , RD _{ID₄})	cost ₄

association procedure, cost-sharing procedure, and routing procedure are explained in the following subsections.

A. Association Procedure

The RD_{FT} or RD_{FT/PT} transmits the network beacon messages periodically to associate with RD_{PT}. The receiving RD_{PT} sends the association request message if the RSSI level of the detected network beacon is above the required RSSI level to achieve the desired packet error rate (PER) for the URLLC communication. The required RSSI level depends on the receiver sensitivity and the quality margin, i.e., the minimum RSSI value above the receiver sensitivity set by the URLLC requirement. The Association Request Message is sent following the Random Access Procedure on the random access time resources announced by the RD_{FT} using uplink transmission. Whereas, the RD_{PT} receives the association response message from the RD_{FT} on the announced downlink time resources as mentioned in [13]. RD_{FT} responds using the Association Response Message IE with a unicast header. The association is completed when the RD_{PT} receives the Association Response Message IE from the RD_{FT} with the positive acknowledgement (ACK). Furthermore, in the association response along with the Association Response Message IE, we propose to multiplex the Cluster Beacon Message and Resource Allocation IE. The Cluster Beacon Message informs the associated RDs about the frame and slot timing, whereas, the Resource Allocation IE informs the associated RDs about their respective scheduled time resources for DL and UL transmissions.

B. Cost Sharing Procedure

In our proposed model, we used uplink transmission exploiting the MAC layer functionality of DECT-2020 NR to share the link information (i.e., latency) of radio devices with the gateway. The MAC layer supports MAC multiplexing header to multiplex multiple MAC SDUs into a single MAC Protocol Data Unit (PDU) as shown in Fig. 2. The MAC multiplexing header includes fields such as MAC Extension (MAC_Ext), IE Type, and Length. The MAC_Ext is a two-bit field which indicates the presence of the Length field and the size of the Length field. The IE type indicates the type of information included in the MAC SDU, e.g., IE type = 001000 indicates the network beacon message and IE type = 001010 indicates the association request message. Lastly, the Length field in the header indicates the length of the MAC SDU (i.e., short SDU, medium SDU, and large SDU). The gateway first calculates the link cost of RDs directly connected with it using the equations mentioned in section

II and saves the link cost with RD_{ID_s} of the connected RDs in the form of a table. Subsequently, $\text{RD}_{\text{FT/PT}}$ shares the link cost with the gateway (i.e., RD_{FT}) whenever there is an association established with other RDs. To share the link cost with the gateway, the $\text{RD}_{\text{FT/PT}}$ adds the cost of the link with RD_{ID} of the two associated devices in MAC SDU. Moreover, we used MAC_Ext and IE type, indicating the length of the IE payload and indicating the presence of the link cost in MAC SDU, respectively. In addition to MAC multiplexing header, the $\text{RD}_{\text{FT/PT}}$ adds the convergence layer unicast header by adding its RD_{ID} as source address and the RD_{ID} of the gateway is set as a destination address. The packet is then forwarded in uplink transmission. In uplink transmission, each intermediate node forwards the packet to the link with minimum cost until the packet reaches the gateway. During the network initialization phase, the gateway receives the packets from the RDs and maintains a table named "DECT-2020 NR edges", which contains RD_{ID_s} and the cost (i.e., latency) of the links. For example consider a small network which contains a gateway, one RD_{FT} , and three RD_{PT_s} with RD_{ID_s} denoted as RD_{G} , RD_{ID_1} , RD_{ID_2} , RD_{ID_3} , and RD_{ID_4} respectively. The stored information at the gateway is shown in Table II.

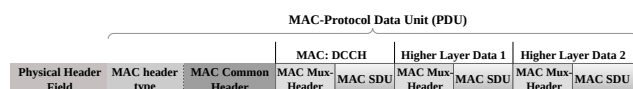


Fig. 2: Physical Header Field and Protocol Data Unit (PDU) of MAC layer.

C. Optimal path selection

The objective of the problem formulation in the section II is to find the optimal path from the gateway to the destination node such that end-to-end latency is minimized. To find the optimal path, the gateway first reads the table "DECT-2020 NR edges" and creates a graph denoted as $G = (V, E)$ where V is the set of vertices i.e., set of DECT-2020 NR radio devices and E is the set of edges i.e., set of links between the radio devices. The cost of the edges is defined using the cost mentioned in the table "DECT-2020 NR edges" i.e., the latency of the links. The gateway then applies Dijkstra's algorithm on graph G to find the optimal path for a specified destination RD. The Dijkstra's algorithm is a well-known algorithm to find the shortest path. The algorithm starts from the source node and it traverses each node of the network once. The algorithm follows the edges with minimum cost and at each traversal, it aggregates the link cost until it reaches the required destination node. The algorithm converges with the optimal path and provides aggregated link cost (i.e., end-to-end Latency). In case of handover, the respective $RD_{FT/PT}$ sends the cost of the newly established link to the gateway. The gateway runs the algorithms again to find the optimal path. However, because of the slow speed of RD_{PT} , the frequency of handovers are not very high. The routing procedure is explained in the following section.

D. Routing Procedure

For downlink transmission (i.e., gateway to RD_{PT}), the gateway first finds the optimal path (i.e., list of nodes) as explained in the section III-C. Let $\Omega = \{G, n_1, n_2, \dots, n_d\}$ be the optimal path where G is the gateway node, n_d is the destination node, and $\{n_1, n_2, \dots\}$ is the set of intermediate nodes. The gateway adds the list of RD_{ID} s of the nodes $\{n_1, n_2, \dots, n_d\}$ in the MAC SDU and adds the MAC multiplexing header to the packet by setting the fields MAC_Ext and IE type indicating the length of MAC SDU and the presence of RD_{ID} s of the optimal path inside the MAC SDU. Finally, the gateway adds the unicast header by setting the RD_{ID} of the first intermediate node as the destination address, and the RD_{ID} of the gateway is set as the source address. The gateway forwards the packet to the first intermediate node using the unicast header. At each intermediate node, the node removes the RD_{ID} of itself and forwards the packet to the next intermediate node until the packet reaches the final destination node. For uplink transmission (i.e., RD_{PT} to gateway), RD_{PT} adds a convergence layer unicast header [2] to the packet with RD_{ID} of the gateway as destination address and RD_{ID} of the node itself as source address. The node forwards the packet to the link with minimum cost in uplink transmission and the packet follows the path with minimum cost at each hop until it reaches the gateway.

IV. PERFORMANCE EVALUATION

Performance evaluation of the proposed approach is conducted by considering a topology with six $RD_{FT,PTs}$, eight RD_{PTs} , and one RD_{FT} for an indoor environment of area 100x140 meters, e.g., a factory hall as shown in Fig. 3. The RD_{FT} and $RD_{FT,PTs}$ are deployed at a static position to provide coverage in the region of interest. Moreover, RD_{FT} is a gateway having a connection to the external network and manages the $RD_{FT,PTs}$. The RD_{PTs} are moving at the speed of up to 2m/s in a pre-defined mobility path represented as dotted lines in Fig. 3. Using the network beacon messages, the association between the RDs is established as explained in the section III-A and represented by solid lines in Fig. 3.

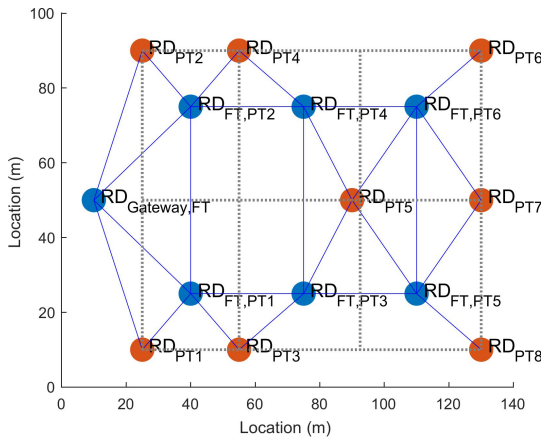


Fig. 3: Network topology of the reference evaluation scenario

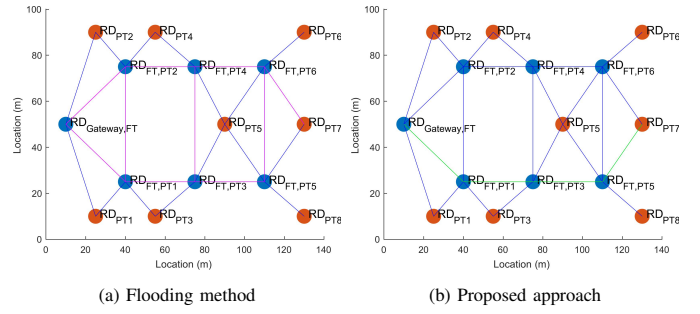


Fig. 4: Path selection using flooding method vs proposed approach

We used four transmission configurations of DECT-2020 NR as shown in Table I to evaluate the proposed solution. Fig. 4 presents the path obtained from gateway to RD_{PT7} using the flooding approach and the proposed solution. The path followed by the flooding method is represented in magenta colour as shown in Fig. 4a. Due to the flooding method, all $RD_{FT,PTs}$ in the network receive the packet at least once and the receiver RD_{PT7} receives multiple copies of the packet, which are discarded using the sequence number embedded in the packet header. Whereas, using the proposed approach, the packet follows the optimal path from the gateway to RD_{PT7} as shown in Fig. 4b

The end-to-end latency of a path denoted as T_{end} is calculated as follows:

$$T_{path} = \sum_{e \in \Omega} T_e \quad (13)$$

$$T_{end} = \frac{\sum_{i=1}^m T_{\Omega}^i}{m} \quad (14)$$

where Ω in equation (13) is the path selected by the flooding method or proposed approach, T_e is the latency of each link of the path calculated using equation (8) or equation (9), and m is number of successful packets received by the destination node. The throughput (in Mbps) of the obtained path is calculated using the equation (15).

$$R = \frac{m \cdot S_p \cdot 8}{(T_{end} - T_{start}) \cdot 10^6} \quad (15)$$

where m is the total number of packets received successfully, S_p is the size of the packet in bytes, T_{start} is the starting time when the first data packet is sent by the gateway, and T_{end} is the time when last packet is successfully received at the destination node.

We calculated the end-to-end Bit Error Rate (BER) [14] for each packet transmission to implement HARQ. If the BER of a received packet is above a threshold [9], the packet is considered corrupt and requires retransmission. Let $v \in \{v_1, v_2, v_3, \dots, V\}$ be the set of nodes/vertices in a path, the end-to-end BER is calculated using the following expression.

$$P_V = (1 - P_{V-1})P(V) + P_{V-1}(1 - P(V)) \quad (16)$$

where P_V is the end-to-end BER, $P(V)$ is the BER of a link/single hop, $(1 - P_{V-1})$ is the probability of correct

bit transmission from the source node to node $V - 1$, and $(1 - P(V))$ is the probability of correct bit transmission from node $V - 1$ to node V . Moreover, the BER is a function of energy spectral density, modulation type, modulation order, and diversity order.

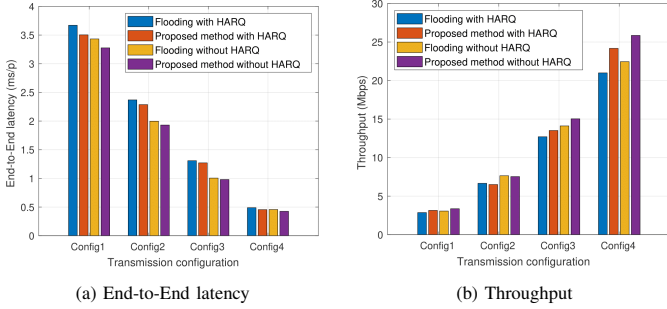


Fig. 5: End-to-End latency and throughput as a function of transmission configuration for RD at static position.

In the next set of experiments, we evaluated our proposed solution with four different transmission configurations (μ, β) as shown in Table I. We used scheduled access for packet transmission, and transport block size for respective transmission configuration (μ, β) for single slot transmission is selected as mentioned in DECT-2020 NR standard [10], [13]. To evaluate the proposed solution, we sent 1000 packets of size 32 bytes for each transmission configuration from gateway to RD_{PT7} (mobile in an area when there are no handovers) and calculated the end-to-end latency and throughput using the equations (14) and (15). From Fig. 5a and Fig. 5b, the effect of transmission configuration (μ, β) on end-to-end latency and throughput is evident due to change in bandwidth, symbol length, subcarrier spacing, and subslot length. It can be observed that the proposed solution outperforms the flooding procedure. The reduction in end-to-end latency and improvement in throughput is observed due to the optimal path selection procedure in comparison to the flooding procedure. The packet routing in flooding is dependent on the time slot scheduling, e.g., in Fig. 3, if RD_{FT,PT2} is scheduled first, the gateway will transmit the packet first to the RD_{FT,PT2} and then in second slot the packet will be transmitted for RD_{FT,PT1}. The time slots are allocated using the first come first serve method, i.e., whenever there is an association between RD_{FT} and RD_{PT} the RD_{FT} allocates time slot for the respective RD. Thereby, the flooding routing procedure is random and unpredictable. Moreover, the flooding approach uses more network resources, increases network congestion, increases latency, is less energy efficient, and uses more bandwidth in comparison with optimal path selection. However, the proposed approach ensures that the packet follows an optimal path and hence utilizes less network resources with minimum end-to-end latency.

In the following experiment, we evaluated the proposed approach with mobile RD_{PT} as a function of transmission configuration. As shown in Fig. 3, RD_{PT} can move on a dotted line with a speed of up to 2m/s in an area of 100x140

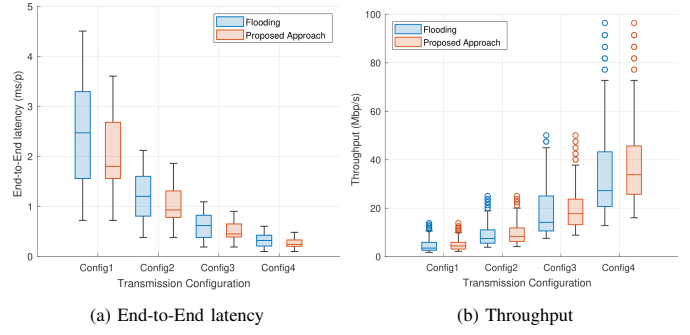


Fig. 6: End-to-End latency and throughput as a function of transmission configuration with mobile RD_{PT} averaged over 1000 iterations.

meters, which means the topology will not change with every second. We evaluated the proposed solution over 1000 iterations, and in each iteration, we considered the location where the handover takes place which changes the topology. For each iteration, we sent 1000 packets and calculated end-to-end latency and throughput for the proposed solution and flooding procedure. It can be observed from the Fig. 6, the proposed solution performed better as compared to the flooding approach. The improved performance of the proposed solution is due to the optimal path selection in comparison to the flooding approach. For the flooding approach, we considered the path through which the first copy of the packet is received successfully while neglecting other paths. However, there are few cases where the proposed solution and flooding approach calculate same end-to-end latency and throughput, e.g., if RD_{PT} is directly connected with the gateway, the latency and throughput are same for both approaches. Including the worst-case scenarios, the improvement of 10% to 15% is observed in end-to-end latency and throughput for various transmission configurations. The percentage of improvement will further increase if there are more intermediate nodes between the gateway and destination node. Since we propose the solution of the use case of URLLC, it can be extended to small to medium-scale networks depending upon the use case. Moreover, due to slow speed of RD_{PT}, the handovers are not very frequent (i.e., every instance), and the overhead of cost sharing is one packet for each handover. Due to handovers, the gateway recalculates the optimal path with time complexity of $(V + E) \log(V)$ which is affordable for small network.

V. CONCLUSION

In this paper, we propose a multi-hop routing mechanism utilizing DECT-2020 New Radio (NR) to support Ultra-Reliable Low Latency Communications (URLLC). The proposed approach involves constructing a network graph at the gateway, where link costs are defined by link-level user plane latency. We introduce a cost-sharing procedure that distributes these link costs to the gateway. The gateway then employs a shortest path optimization algorithm (i.e., Dijkstra's algorithm) to determine the optimal route to the destination node. For downlink multi-hop packet forwarding, the routing

information, represented as a list of RD_{IDs} corresponding to the optimal path, is multiplexed in the MAC Service Data Unit (SDU). This enables intermediate nodes to forward packets to the next hop without relying on traditional flooding techniques. Our solution exhibits a notable improvement in both end-to-end latency and throughput. Future work will focus on implementing and evaluating the proposed method using software-defined radio (SDR) platforms.

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